

Procurement Spend Analysis

Transforming procurement data into actionable insights through Python, PostgreSQL, and Power BI to optimize spending and supplier relationships.



End-to-End Analytics Pipeline

01

Data Preparation

Python-based data cleaning and exploratory analysis to understand spending patterns and identify quality issues.

02

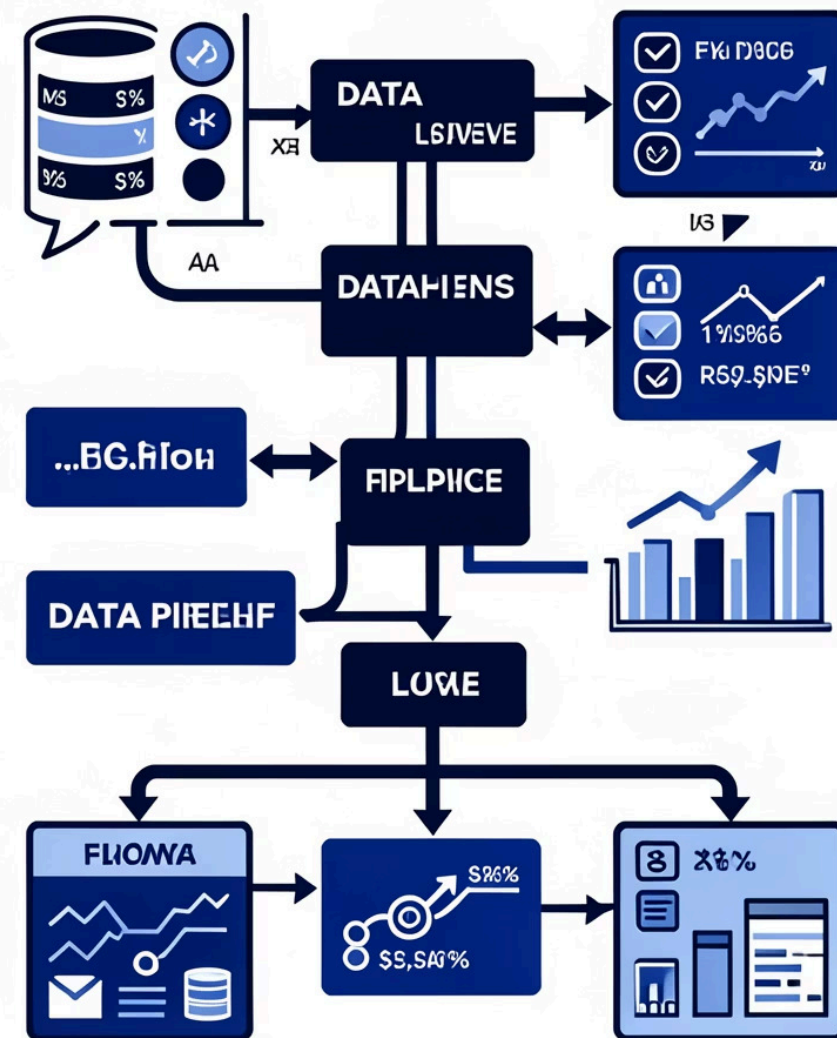
Structured Querying

PostgreSQL database design enabling scalable analysis of supplier dependency and pricing inefficiencies.

03

Interactive Visualization

Power BI dashboards delivering business insights and cost optimization opportunities to stakeholders.



Dataset Overview

500

Total Records

Complete dataset with no missing or duplicate entries

9

Key Attributes

Transaction details, cost metrics, and procurement context

Key Attributes

- Transaction Details: ID, Item Name, Category
- Cost Metrics: Quantity, Unit Price, Total Cost
- Procurement Context: Supplier, Buyer, Purchase Date

Data					
ER	ST	FL	FA	So	PALT
\$,900					\$,600
\$,604	\$80	355	450		\$100
\$,900	\$30	300	450		\$700
\$,600	\$50	50	550		\$,600
\$,104	\$100	330	\$40		\$700
\$,300	\$30	\$30	\$20		\$300
4,500	\$100	810	\$30		\$700
4,160	\$80	310	330		\$,600
\$600	\$150	330	530		\$500

Data Quality & Exploratory Insights

Data Validation

Dataset verified as complete with no missing records. Data types corrected and purchase dates standardized to datetime format.

Distribution Patterns

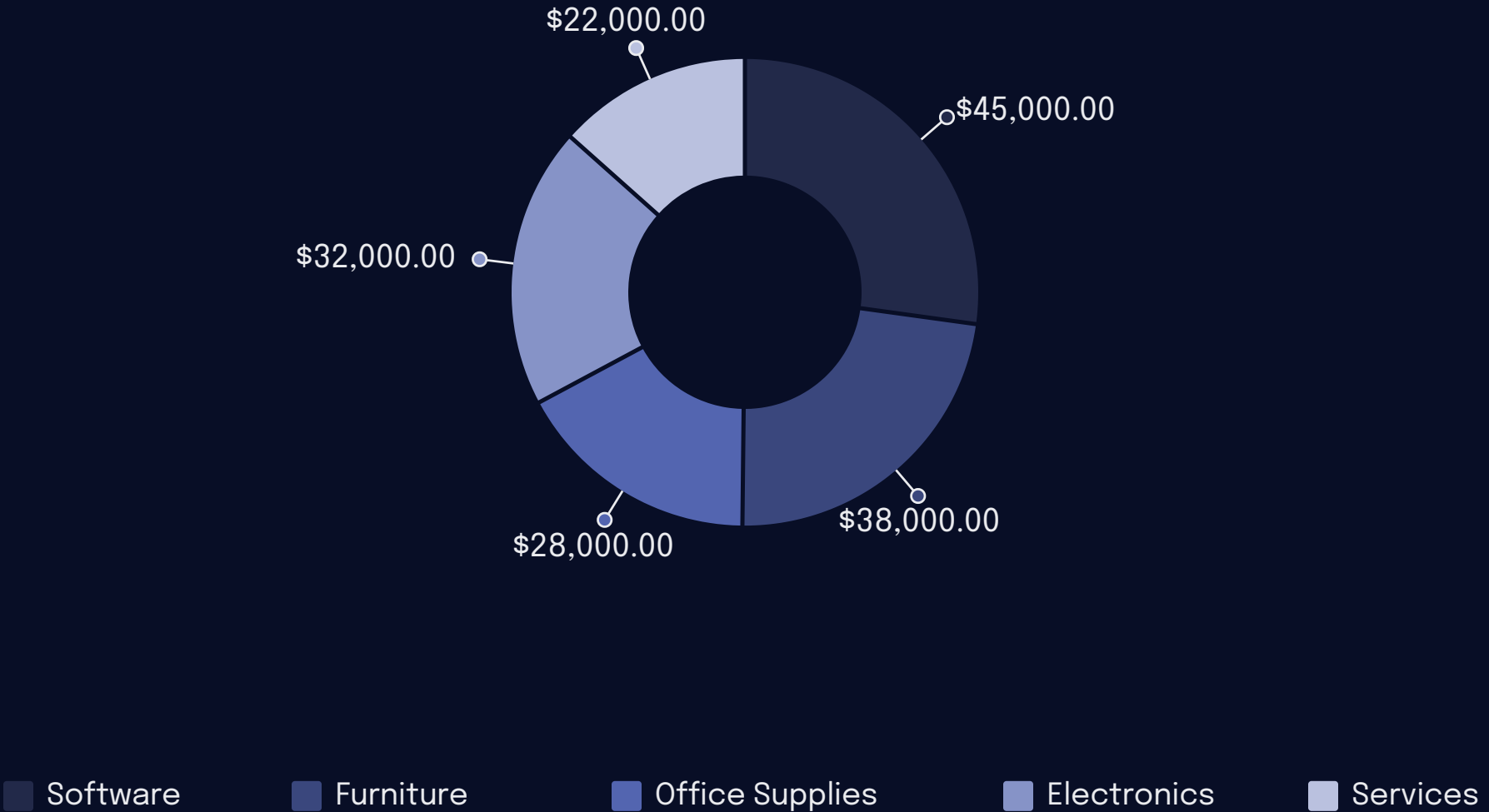
Transaction costs are right-skewed, indicating a small number of high-value purchases drive overall spend.

Category Dominance

Strategic categories like software and furniture dominate spending, with significant buyer-level variation.

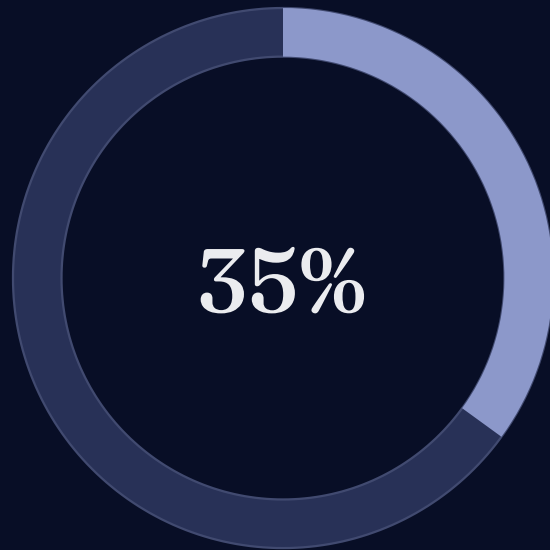
Total Procurement Exposure

Spend is heavily concentrated in a few categories, indicating targeted cost control opportunities.

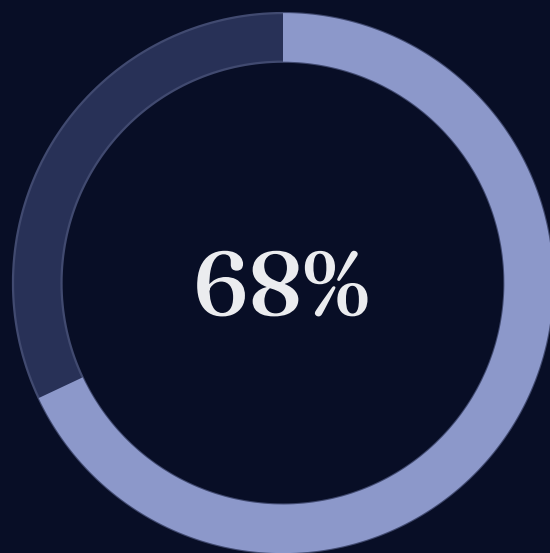


Supplier Dependency Risk

A small number of suppliers account for a large share of total spend, creating significant dependency risk and negotiation leverage concerns.



Top supplier spend share



Top 3 suppliers combined



High-Value Transaction Insights

Transaction Volume

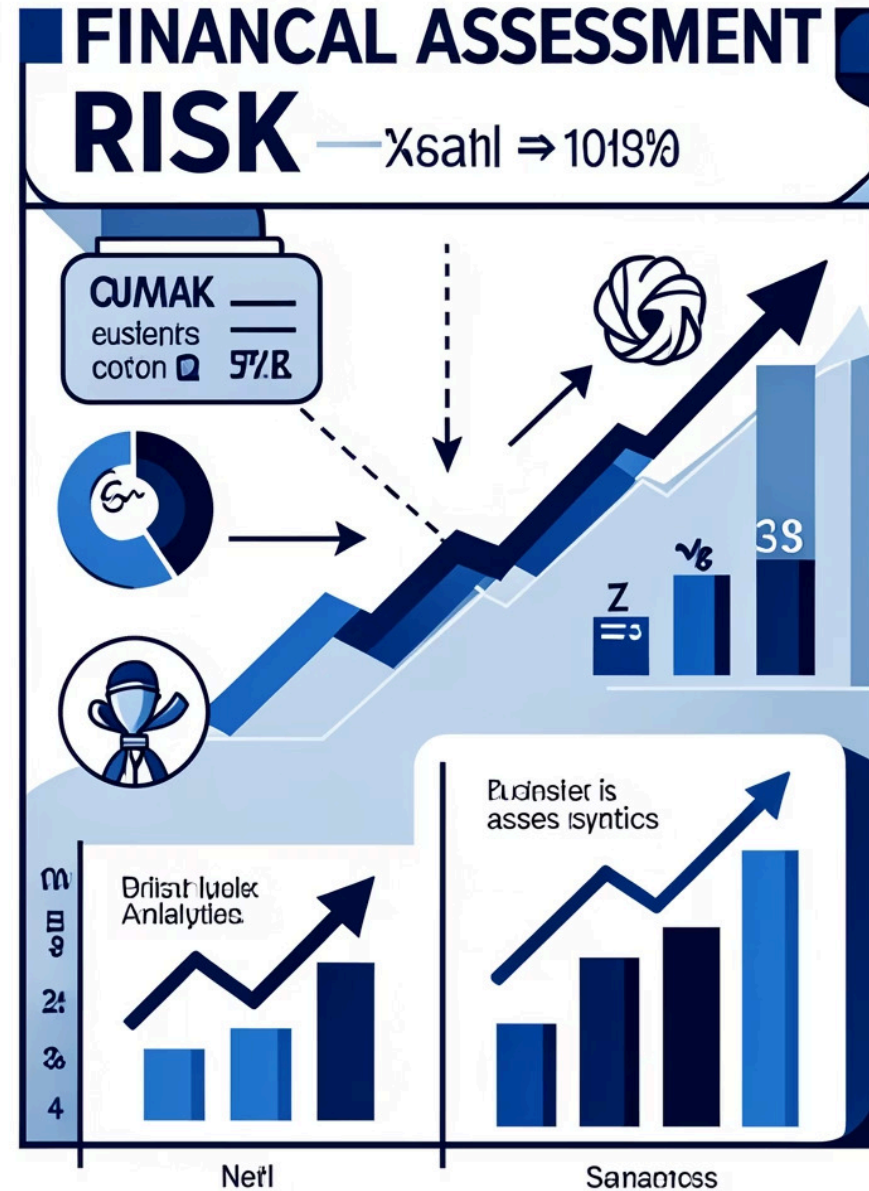
500 total transactions with average value requiring baseline approval controls.

High-Risk Threshold

Transactions exceeding \$5,000 are infrequent but significantly impact total spend.

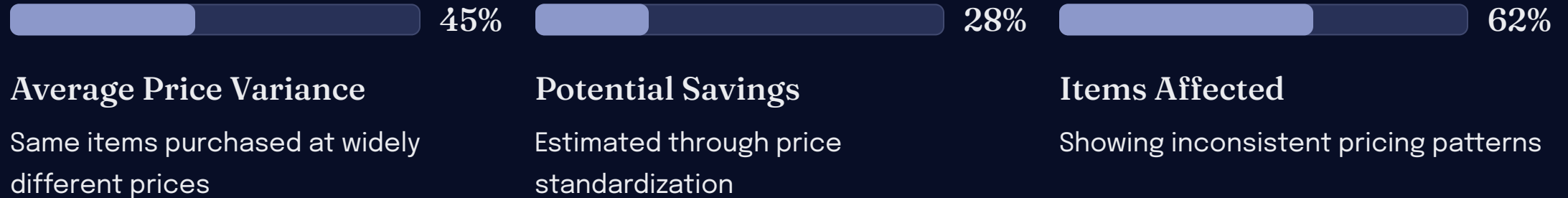
Control Requirement

High-value purchases require stricter approval workflows and governance oversight.



Price Inconsistency Analysis

Significant price variations exist for identical items across purchases, revealing lack of standardization and negotiation gaps.





Strategic Recommendations



Governance Controls

Introduce approval thresholds for high-value purchases to manage financial risk.



Supplier Consolidation

Consolidate suppliers for strategic categories to reduce dependency and improve negotiation power.



Price Standardization

Establish pricing benchmarks and monitor supplier efficiency regularly for cost optimization.



Future Roadmap

Predictive Analytics

Cost forecasting and trend prediction

Real-Time Integration

Live BI dashboards and alerts

1

2

3

4

Performance Scoring

Automated supplier evaluation metrics

Anomaly Detection

AI-powered spend monitoring

This project demonstrates a complete procurement analytics pipeline, transforming raw transactional data into actionable insights that support strategic decision-making and identify optimization opportunities.